UU-P-154c
NOVEMBER 30, 1960

SUPERSEDING
Int. Fed. Spec. UU-P-00154b(GSA-FAS)
July 29, 1960 and
Fed. Spec. UU-P-154a
April 3, 1960

FEDERAL SPECIFICATION

PAPER, CARBON, PENCIL

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope.—This specification covers carbon paper for pencil use (see 6.1).

1.2 Classification.

1.2.1 *Types.*—The carbon paper shall be furnished in the following types, as specified in the invitation for bids, contract, or order (see 6.2):

Type I. —One-time use.
Type II. —Multiple use.
Type III.—Multiple copy.

1.2.2 *Sizes and colors.*—The carbon paper shall be furnished in sizes and colors as specified in the invitation for bids, contract, or order (see 6.2).

1.2.3 *Coating* (see 3.1).

2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:

UU-P-31 — Paper; General Specifications and Methods of Testing.
UU-P-121—Paper; Bond, and Writing, White and Colored.
UU-P-328—Paper, Manifold.
PPP-B-585—Boxes, Wood, Wirebound.
PPP-B-591—Boxes, Fiberboard, Wood-Cleated.
PPP-B-601—Boxes, Wood, Cleated-Plywood.
PPP-B-621—Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636—Box, Fiberboard.
PPP-B-676 — Boxes, Set-Up, Paperboard.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.
Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

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and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Specifications:

MIL–B–10377—Box, Wood, Cleated, Veneer, Paper Overlaid.

MIL–L–10547 — Liners, Case, Waterproof.

Military Standards:

MIL–STD–105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL–STD–129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials.—The carbon paper shall consist of carbonizing paper coated on one or two sides, as specified (see 6.2), with a suitable form of carbon and other materials necessary to give the required copying qualities as specified herein.

3.1.1 *Paper stock.*—The carbonizing paper stock shall be free from groundwood pulp when tested as specified in 4.4.1.

3.2 Performance.

3.2.1 *Resistance to curl.* — Carbon paper shall not curl more than 20 percent when tested as specified in 4.4.2.

3.2.2 *Serviceability and manifolding.*

3.2.2.1 *Serviceability.* — When tested as specified in 4.4.3.1, the impressions made from the same carbon-paper area shall be examined and the number of legible impressions shall be as specified in table I. The last impression shall be clearly and easily legible due to carbon transfer.

3.2.2.2 *Manifolding.*—When tested as specified in 4.4.3.2, the impression shall be clearly legible on the number of copy sheets specified in table I. All copies of the required manifolding set shall be clearly and easily legible due to carbon transfer.

TABLE I.—*Serviceability and manifolding*

	Type I	Type II	Type III
Serviceability: Minimum number of legible impressions	--	15	10
Manifolding: Minimum number of legible copies	2	2	6

3.3 Pinholes.—The carbon paper shall be free from pinholes having a 0.006- to 0.007-inch diameter when tested as specified in 4.4.4.

3.4 Workmanship. — The carbon paper shall be free from lumps or knots of fibers. The coating shall be smooth to the touch and free from holes. It shall uniformly cover the entire surface of one or both sides of the tissue, as ordered (see 6.2) and shall not flake during any of the tests specified in section 4. The occurrence of defects shall not exceed the acceptable quality levels stipulated in the quality assurance provisions.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Supplier's responsibility.—The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed neces-

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sary to assure suppplies and services conform to prescribed requirements.

4.2 **Lot.**—A lot of carbon paper shall consist of all carbon paper of the same type, size, color, and coating offered for delivery at one time.

4.3 **Sampling.**

4.3.1 *Sampling of shipping containers.*—The number of shipping containers taken for examination from each lot offered for delivery shall be in accordance with Level I, table III, of Military Standard MIL–STD–105. The sample shall be examined for presence of defects listed in table II below. The Acceptable Quality Level (AQL) shall be 4.0 percent defective.

TABLE II.—*Shipping container defects*

Examine	Defects
Shipping container	Type, class, style, grade, and closure not in compliance with applicable specifications referenced in section 5.
Box maker's certificate	Burst test not as specified.
Quantity of boxes in shipping container	Not as specified.
Arrangement of boxes in shipping container	Arrangement and placement of boxes not as specified. Boxes not snugly packed.
Shipping container marking	Missing, illegible, incorrect, or incomplete.

4.3.2 *Sampling of unit boxes.*—Unit boxes of carbon paper to be examined for packaging and marking/labeling defects shall be taken from shipping containers taken in accordance with 4.3.1. The sample boxes shall be examined for the presence of packaging and marking/labeling defects listed in table III, except that one box shall be retained unopened for the curl-resistance test specified in 4.4.2. The sample size for boxes shall be in accordance with Military Standard MIL–STD–105, Inspection Level L–4; however, not more than three boxes shall be taken from any one shipping container. The AQL shall be 4.0 percent defective.

TABLE III.—*Unit box defects*

Examine	Defects
Unit box	Type, class, and style not in compliance with Federal Specification PPP–B–676.
Box top	Inadequately fabricated to provide end opening hinged cover.
Construction of box (if approved equivalent box is furnished in lieu of specification box)	Not constructed of materials equal in strength to specification box.
Cover-wrap of box	Cover-wrap not as specified (where applicable).
Arrangement of folders in unit boxes	Slippage of papers and/or folders within the box. A tight fit not provided. Cardboard stiffener missing (where applicable).
Folders	Not as specified.
Count:	
Average number of sheets per box	Less than the number specified.
Number of sheets per folder	Less than required amount by more than 2.
Number of folders per box	Less than specified number.
Unit box marking or labeling	Missing, illegible, incorrect, or incomplete.

4.3.3 *Sampling for sheet examination.*—Sample sheets of carbon paper shall be taken from the boxes examined in accordance with 4.3.2. The unit of product shall be one box of carbon paper. Not less than 5 sheets shall be taken from each box and examined for the applicable defects listed in table IV, except that

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the total number of sheets shall not exceed 225. The AQL shall be 10.0 defects per hundred units.

TABLE IV.—*Carbon paper defects*

Examine	Defects
Wrinkles and waves	Any wrinkle or wave which can be felt when carbon paper is laid on a flat surface uncarbonized-side up and fingers are run over the surface.
Appearance	Carbon-coated side not readily identifiable. Any carbon coating material on back of sheet. Any stain (coating material) on back of any sheet visible when viewed by reflected light at a distance of 18 inches.
Uniformity of coating (Hold so that bright light strikes back of sheet)	Any area coated so lightly as to be translucent. Light lines and non-uniform appearance of coating. Flaking.
Dimensions (size)	Varies from specified size by more than plus ⅛ inch or minus 1/16 inch in any direction.
Holes, etc.	Any hole visible when sheet is laid out flat on a white opaque surface carbon side up.
Workmanship	Any tear or cut; any crease. Any crimped corner or side; lump or knot of fiber.

4.3.4 *Sampling for tests.*—Sample sheets of carbon paper shall be taken for tests from the samples selected in accordance with 4.3.3. The unit of product shall be one box of carbon paper. Samples, sufficient for tests, shall be taken from not less than 5 boxes of carbon paper. The sample sheets shall be tested for the characteristics specified in table V. Failure to meet requirement for any test characteristic shall be cause for rejection of the lot.

4.4 Tests.

4.4.1 *Stock.*—The composition of the stock shall be determined by method 300 of Federal Specification UU–P–31.

4.4.2 *Curl-resistance test.*—Two sheets of the carbon paper to be tested shall be removed from a previously unopened box and placed on a flat surface, one sheet with carbon side up and the other sheet with carbon side down. The sheets shall be exposed for 1 minute under atmospheric conditions of 50 ± 2 percent relative humidity and 23° ± 2°C. (73° ± 3.5° F.). The sum of the lengths of the portions of the short sides on each sheet not in contact with the flat surface divided by the sum of the length of the narrow sides multiplied by 100 equals the percent curl.

4.4.3 *Serviceability and manifolding test.*

4.4.3.1 *Serviceability.*—The test shall be performed on a specimen of the carbon paper measuring 2 by 6 inches. The test piece shall be laid carbon side down on a glass plate with its long dimension parallel to one side of the plate. Two pieces of 20-pound bond paper conforming to type III of Federal Specification UU–P–121, measuring 4 by 10½ inches, shall be superimposed one on the other; the 4-inch edges shall be joined at one end with a fold of tape. The joined sheets shall be placed one under and one over the carbon-paper test sample, their long sides being perpendicular to the long side of the test sample and the joined edges extending 1 inch beyond the test sample. The test sample shall be secured to the glass plate at the 2-inch ends with tape. A straightedge shall be placed over the test assembly parallel to the 6-inch edge of the sample at the midpoint of the sample and shall be secured to the glass plate. A fine-point, ball point pen producing a line width impression of 0.37 to 0.39 millimeter held in a vertical position under a pressure of 165 grams shall be used to make impressions (see 6.3). The glass

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TABLE V.—*Carbon paper testing requirements*

Characteristic	Requirement reference	Test paragraph reference	Requirement applicable to—		Number of determinations per sample unit	Results reported as—	
			Sample unit	Lot average		Pass or fail	Numerically to nearest—
Paper stock	3.1.1	4.4.1	Composite		1	X	----
Resistance to curl	3.2.1	4.4.2		X	1		1.0 percent
Pinholes	3.3	4.4.4	X		1	X	----
Serviceability (wear-down)	3.2.2.1, table I	4.4.3.1	X		2	X	----
Manifolding	3.2.2.2, table I	4.4.3.2	X		2	X	----
Flaking of coating	3.4	4.4 through 4.4.3.2	X		--	X	----

plate to which the test assembly has been secured shall be laid on a drawing board, and the ball point under pressure placed over the assembly so that it is in contact with the straightedge. A 4-inch long impression shall be made by pulling the assembly slowly in a straight line, using the straightedge as a guide. The joined bond sheets shall then be withdrawn approximately ¼ inch to expose an unmarked area on the impression sheet and another impression made by pulling the assembly over exactly the same path along the straightedge. The operation shall be repeated until 20 impressions have been made on the copy sheet. The number of impressions specified in table I shall be inspected relative to legibility. The carbon paper shall be inspected for evidence of flaking.

4.4.3.2 *Manifolding.*—The number of manifold copies that the carbon paper will make at one time shall be determined by drawing a test packet assembly under a ball point pen at a pressure of 165 grams as in 4.4.3.1. The test packet shall consist of a first sheet of 20-pound bond paper conforming to type III of Federal Specification UU–P–121, 6 copy sheets of manifold paper conforming to type IV of Federal Specification UU–P–328, and 6 sheets of the carbon paper. The sheets shall measure 2 by 4 inches. The first sheet and copy sheets shall be stapled together at each 2-inch end to form a smooth packet. The packet shall be secured to a glass plate with cellulosic adhesive tape. A straightedge shall be attached as specified in 4.4.3.1. Three parallel lines shall be made by drawing the assembly slowly under the pen. The straightedge shall be moved ¼ inch to a previously unmarked area after each impression is made. The manifold impressions shall be examined for legibility and the carbon paper examined for flaking.

4.4.4 *Pinholes.*—Twenty-five square inches on each of twelve sheets (300 square inches) taken at random from the boxes drawn for sampling as specified in 4.3.3 shall be examined for freedom from pinholes. Care should be exercised when a comparison hole is made by puncturing a sheet of carbon paper with a wire 0.006 to 0.007 inch in diameter that the hole is not made larger than the diameter of the wire. Examination of carbon paper for the presence of pinholes shall be made in a dark room. Either a 20-watt frosted-glass

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fluorescent tube or a 7½-watt white bright-light bulb may be used to provide the light source. A template having a 5- by 5-inch opening (prepared from cardboard or other like material) may be used to define the area of the paper to be examined. A source of wire 0.006 to 0.007 inch in diameter may be found in some multiple-strand copper electrical extension cords. The wire shall be measured with a micrometer to assure that it is the proper diameter. In examining the carbon paper, place the specimen with the carbon side up on the glass plate directly over the light. Place the template over the carbon paper to define the area to be examined.

5. PREPARATION FOR DELIVERY

Federal Standard No. 102 (for civil agencies only) should be referred to for definitions and applications of the various levels of packaging and packing for civilian procurements.

5.1 Packaging.—Packaging shall be level A, B, or C as specified (see 6.2).

5.1.1 *Levels A and B.*—Carbon paper of one type, size, color, and coating shall be packaged 25 sheets to a paper folder, 100 sheets (4 folders) to a set-up paperboard box conforming to type I, class A, style 4 of Federal Specification PPP–B–676. As alternate construction to the unhinged box covered by Federal Specification PPP–B–676, the top of the box may be hinged at one end or side or scored or perforated for folding back at a point approximately midway of its longer dimension. Such alternate construction shall be such that the box will meet the quality requirements of the specification. Boxes hinged in the center shall have that portion of the box top above the hinge securely attached to the bottom of the box. All boxes shall fit the contents snugly so that no slippage of paper or folders will occur during handling or shipment. As necessary, a full length sheet of chipboard, paperboard, or fiberboard filler shall be placed on top of the folders within the box to fill excess space. Each box shall be secured in a manner to prevent accidental opening unless the covers are sufficiently tight fitting to preclude accidental opening.

5.1.2 *Level C.*—One hundred sheets of carbon paper shall be packaged in a box in accordance with the manufacturer's standard commercial practice.

5.2 Packing.—Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 *Level A.* — Fifty boxes of carbon paper where neither dimension of the sheets exceeds 14½ inches; or twenty-five boxes where one or both dimensions exceed 14½ inches, of one type, size, color, and coating packaged as specified in 5.1, shall be packed in a snug-fitting shipping container conforming to class 2, V3c or V3s, style RSC or CSSC of Federal Specification PPP–B–636. Alternatively, shipping containers may be used that conform to overseas-type of Federal Specification PPP–B–591, PPP–B–601, or Military Specification MIL–B–10377; class 2, style 2 or 4 of Federal Specification PPP–B–621; or style 1, 2 or 2A, class 3 use, type 2 load of Federal Specification PPP–B–585. The boxes shall be packed flat in two stacks of 25 boxes each, placed side by side with the longest dimension adjacent where packing requires 50 boxes; or a single stack of 25 boxes where packing requires 25 boxes only. Each shipping container shall be provided with a type I or II, grade B or C, class 2 sealed case liner conforming to Military Specification MIL–L–10547. Each shipping container shall be closed and reinforced with steel strapping or tape banding in accordance with the appendix to the applicable container specification.

5.2.2 *Level B.*—Fifty boxes of carbon paper where neither dimension of the sheets exceeds 14½ inches; or twenty-five boxes where one or both dimensions exceed 14½ inches, of one type, size, color and coating packaged as specified in 5.1, shall be packed in a snug-fitting shipping container conforming to type I or II, class 1, style RSC or CSSC of Federal Specification PPP–B–636, except that the minimum bursting strength of the fiberboard

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shall be 275 p.s.i. All flaps of the fiberboard shipping containers shall be securely sealed with an adhesive commercially used for the specific product being packed by application throughout the entire area of contact between the flaps, or by a combination of metal stitching the bottom flaps and sealing the top flaps with adhesive. Alternatively, shipping containers may be used that conform to domestic-type, style A or B of Federal Specification PPP–B–591, PPP–B–601, or Military Specification MIL–B–10377; class 1, style 2 or 4 of Federal Specification PPP–B–621; or style 1, 2 or 2A, class 1 use, type 2 load of Federal Specification PPP–B–585. The boxes shall be packed flat in two stacks of 25 boxes each, placed side by side with the longest dimension adjacent where packing requires 50 boxes; or a single stack of 25 boxes where packing requires 25 boxes only.

5.2.3 *Level C.*—Fifty boxes of carbon paper where neither dimension of the sheets exceeds 14½ inches; or twenty-five boxes where one or both dimensions exceed 14½ inches, of one type, size, color and coating packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination. Containers shall conform to the carrier rules and regulations applicable to the mode of transportation.

5.3 Marking.

5.3.1 *Civil agencies.* — Unless otherwise specified, each box of 100 sheets of carbon paper shall be marked or labeled on the end which folds back with applicable Federal stock number, item identification, including specification classification (such as type, size, color, coating, etc.), quantity of sheets contained, and applicable contract or order number. Shipping containers shall be marked in accordance with Federal Standard No. 123, including specification classification. Additionally, the four side panels of the shipping container shall be marked with arrows pointing to the top of the container and the word "UP" in letters of 1-inch minimum height centered on each panel near top edge.

5.3.2 *Military agencies.*—In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with Military Standard MIL–STD–129.

6. NOTES

6.1 Intended use.—The carbon paper is intended for making carbon copies, using a pencil to make the original copy; type I, for one time use only; type II, for repeated use; and type III, for manifolding with repeated use.

6.2 Ordering data. — Purchasers should specify the following:

(*a*) Title, number and date of this specification.
(*b*) Type (see 1.2.1).
(*c*) Size and color (see 1.2.2).
(*d*) Whether carbon paper shall be coated one or two sides (see 3.1).
(*e*) Applicable levels of packaging and packing and any special marking required (see 5.1, 5.2, and 5.3).

6.3 The use of a penetrometer (Universal Model 570, method 311.5, Federal Test Method Standard No. 791, Lubricants, Liquid Fuels and Related Products; Methods of Testing.) is suggested for holding the pen in a vertical position under known pressure. A metal ball point refill cartridge which has been shortened to 1 inch is used in place of the penetration cone. The solenoid mechanism is pushed in and secured to allow free vertical movement of the plunger. A 150 gram load is placed on the 15 gm. aluminum plunger for an overall total weight of 165 grams.

6.4 Military standards. — Items procured under this specification for military use are to be limited to the variety shown on the applicable military standards. Personnel of the military departments are requested to refer to these documents for guidance.

6.5 Transportation data.—Transportation descriptions and minimum weights applicable to this commodity are:

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Type I:

Rail

One-time carbon paper.

Carload minimum weight 36,000 pounds.

Motor

One-time carbon paper.

Motor volume minimum weight 36,000 pounds.

Types II and III:

Rail

Paper, carbon, not otherwise indexed by name.

Carload minimum weight 36,000 pounds.

Motor

Paper, carbon, not otherwise indexed.

Truckload minimum weight (w) 36.6 pounds, (w) subject to Rule 34, National Motor Freight Classification.

Notice. — When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY CUSTODIANS:

Army—Quartermaster Corps
Navy—Bureau of Supplies and Accounts
Air Force—Mobile Air Materiel Area.

Copies of this specification may be purchased for 5 cents each.

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